Homework 7
Numerical Methods
Fall 2025

Due Tuesday Dec 2

Instructions:

Complete the tasks given below. Provide a summary of your findings and results.

1 Derivative approximation for a single variable function $u = u(x)$

Perform Taylor Series expansions to derive formulas for the following derivatives and determine the order of the approximation:

a) $u'(x_i)$ using two points x_i, x_{i+1}
b) $u'(x_i)$ using two points x_{i-1}, x_i
c) $u'(x_i)$ using two points x_{i-1}, x_{i+1}
d) $u'(x_i)$ using four points $x_i, x_{i+1}, x_{i+2}, x_{i+3}$

2 Numerical Solution to ODEs

Consider the linear second-order boundary value problem

$$u''(x) + P(x)u'(x) + Q(x)u = f(x), \quad u(a) = \alpha, \quad u(b) = \beta$$

where $a = x_0 < x_1 < x_2 < ... < x_{N-1} < x_N = b$ represents an equally spaced partition of the interval $[a, b]$ - that is $x_i = a + ih$, where $i = 0, 1, 2, ..., N$ and $h = (b - a)/N$. The interior mesh points are

$$x_1 = a + h, x_2 = a + 2h, ..., x_{N-1} = a + (N - 1)h.$$

Letting $u_i = u(x_i), P_i = P(x_i), Q_i = Q(x_i),$ and $f_i = f(x_i)$ and if both $u''(x)$ and $u'(x)$ are approximated with central differences (to match 2nd order accuracy) we get

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + P_i \frac{u_{i+1} - u_{i-1}}{2h} + Q_i u_i = f_i \tag{*}$$

which can be simplified to

$$(1 + \frac{h}{2}P_i)u_{i+1} + (-2 + h^2 Q_i)u_i + (1 - \frac{h}{2}P_i)u_{i-1} = h^2 f_i \tag{**}$$

Use $(*)$ or $(**)$ to obtain a numerical approximation for $u(x)$ for the following problem:

$$u''(x) + 3u'(x) + 2u(x) = 4x^2, \quad u(1) = 1, \quad u(2) = 6$$

a) Use $N = 10$ to plot the approximation, $u(x)$ vs x.

b) Use $N = 40$ to plot the approximation along with the exact solution $u_{exact}(x)$.

Note:

$$u_{exact}(x) = 7 - 6x + 2x^2 + C_1 e^{-2x} + C_2 e^{-x}$$

where

$$C_1 = -\frac{e^3(2 + 3e)}{e - 1}, \quad C_2 = \frac{2e + 3e^3}{e - 1}.$$

3 Numerical Integration - and Richardson Extrapolation

a) Determine the number of subintervals N needed to approximate

$$\int_0^\pi \sin(x)dx$$

with an error not to exceed 10^{-4} using the composite trapezoid rule.

b) Using the result found in a), compute the actual error. Discuss your findings.

c) Richardson Extrapolation: recall that this procedure was used to improve the error in the approximation of the second order scheme, $\mathcal{O}(h^2)$, given by

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

Setting $\phi(h)$ as the approximation to $f'(x)$, we showed that

$$\phi(h) = \frac{4\phi(\frac{h}{2}) - \phi(h)}{3}$$

defines a fourth order approximation $\mathcal{O}(h^4)$ for $f'(x)$. The scheme tells you that if you approximate the first derivative with $\phi(h)$ and $\phi(h/2)$ then by performing a simple linear combination of your approximations will redefine your derivative approximation at an increased order of accuracy.

Your Task: Starting from

$$f'(x) = \frac{f(x+h) - f(x)}{h}$$

which gives a first order, $\mathcal{O}(h)$, approximation, increase the accuracy by means of applying (1 time only) Richardson Extrapolation to your first derivative formula. Express the form of your formula and include the error term. What order do you get?